

Software Requirements Specification (SRS)

Project: Voice Assistant for Multimedia Control

Version: 1.0

Authors: (TEAM 10) Yash Raghavendra A (PES2UG24CS913)

Kushi Niranjana (PES2UG24CS245)

Hemant Ram Nelapati (PES2UG24CS192)

Keerthi B Patil (PES2UG24CS225)

Date: 05-09-2026

Status: Final for Submission

Revision history

Version	Date	Author	Change summary	Approval
1.0	05-09-2026	Project Team	SRS for Voice Assistant for Multimedia Control with diagrams embedded	

Approvals

Role	Name	Signature / Email	Date
Course Coordinator			

TABLE OF CONTENTS

Contents

1. Introduction.....	3
1.1 Purpose	3
1.2 Scope.....	3
1.3 Audience.....	3
1.4 Definitions	3
2. Overall Description.....	4
2.1 Product Perspective	4
2.2 Major Product Functions (Detailed)	4
2.3 User Roles and Characteristics (Expanded)	4
2.4 Operating Environment	4
2.5 Constraints	4
3. External Interface Requirements	5
3.1 User Interfaces	5
3.2 Hardware Interfaces.....	5
3.3 Software Interfaces	5
3.4 Communications.....	5
4. System Features (Detailed).....	6
5. Non-Functional Requirements (Detailed).....	8
5.1 Security.....	9
5.1.1 Security Objectives.....	9
5.1.2 Security Requirements.....	9
6. Quality Attributes & Acceptance Tests	11
7. UML Use-Case Diagram	12
7.1 Use-Case Diagram - Voice Command to Media Control.....	12
7.2 Use-Case Diagram - Multimedia Operations	12
8. Requirements Traceability Matrix (RTM).....	13

1. Introduction

1.1 Purpose

This Software Requirements Specification (SRS) defines the requirements, interfaces, security objectives, verification criteria, UML use cases, and traceability for a Raspberry Pi-based Voice Assistant for Multimedia Controls.

1.2 Scope

The system captures spoken commands through a USB microphone connected to the Raspberry Pi, converts speech to text using a speech-recognition engine, identifies the intended multimedia action, validates the command, and controls a media player such as VLC. Supported controls include play, pause, resume, stop, next, previous, volume up, volume down, and mute. Gesture recognition, camera input, and MediaPipe are explicitly outside the scope of this voice-only version.

1.3 Audience

Developers, QA/test engineers, course instructors, project evaluators, and project team members.

1.4 Definitions

SRS – Software Requirements Specification; STT/ASR – Speech-to-Text / Automatic Speech Recognition; UI – User Interface; VLC – VideoLAN media player; Intent – normalized action derived from a spoken command; Pi – Raspberry Pi.

2. Overall Description

2.1 Product Perspective

The Raspberry Pi is the central processing and control unit. A USB microphone provides voice input. The speech-recognition component converts speech to text, a Python command processor maps recognized text to predefined intents, a validator checks the intent, and a media controller operates VLC/media playback. Audio is delivered through an output device connected to the Raspberry Pi.

2.2 Major Product Functions

- Capture voice commands.
- Convert speech to text.
- Detect supported multimedia intents.
- Validate commands using an allow-list.
- Play, pause, resume, and stop media.
- Select next and previous media items.
- Increase, decrease, and mute volume.
- Provide command/error feedback.
- Record timestamped command/result events without raw audio by default.
- Recover safely from recognition and media-player errors.

2.3 User Roles and Characteristics

End User: gives spoken multimedia commands and expects simple, quick responses.

Project Developer: configures the Pi, recognition engine, parser, and media controller.

Tester/Evaluator: verifies functionality, performance, security, and acceptance criteria.

2.4 Operating Environment

Raspberry Pi OS/Linux environment, Python 3, USB microphone, VLC or compatible media player, and an audio output device. Network connectivity depends on the selected speech-recognition engine; an offline engine may be used where supported.

2.5 Constraints

- Recognition quality depends on microphone quality and ambient noise.
- Only predefined multimedia intents may be executed.
- Raw voice recordings shall not be stored during normal operation.

3. External Interface Requirements

3.1 User Interfaces

The primary interface is voice. A terminal or lightweight UI may display recognized text, detected intent, execution status, and errors.

3.2 Hardware Interfaces

- USB microphone connected to Raspberry Pi.
- Audio output connected to Raspberry Pi through a supported audio path.
- Optional status LED/button for demonstration or activation.

3.3 Software Interfaces

- Speech-recognition engine/library.
- Python command-processing and intent-detection module.
- VLC/media-player control interface.
- Local logging component.

3.4 Communications

If the selected speech-recognition engine uses network communication, its supported secure transport shall be used. Core command processing and media control shall remain local to the Raspberry Pi where practical. Network failure shall not cause unintended multimedia actions.

The project specifies 18 functional requirements, 7 non-functional requirements, 3 security objectives, and 6 security requirements.

4. System Features

Each requirement includes acceptance criteria and a reference test case. IDs follow VA-F-####.

Functional Requirements

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case
VA-F-001	The system shall capture voice input from the USB microphone.	Functional	High	End User	Captured audio available for a test command; TC-Voice-01
VA-F-002	The system shall convert captured speech into text.	Functional	High	End User / Developer	Known phrases produce recognized text; TC-Voice-02
VA-F-003	The system shall detect the intent of a recognized multimedia command.	Functional	High	Developer	Supported phrase maps to correct intent; TC-Intent-01
VA-F-004	The system shall validate detected intents against a supported-command allow-list.	Functional	High	Security / Developer	Unsupported intent causes no action; TC-Sec-03
VA-F-005	The system shall play media for a play command.	Functional	High	End User	PLAY starts playback; TC-Media-01
VA-F-006	The system shall pause current media for a pause command.	Functional	High	End User	PAUSE retains playback position; TC-Media-02
VA-F-007	The system shall resume paused media for a resume command.	Functional	High	End User	RESUME continues playback; TC-Media-03
VA-F-008	The system shall stop current media for a stop command.	Functional	High	End User	STOP ends playback; TC-Media-04
VA-F-009	The system shall select the next media item for a next command.	Functional	Medium	End User	NEXT selects next available item; TC-Media-05
VA-F-010	The system shall select the previous media item for a previous command.	Functional	Medium	End User	PREVIOUS selects previous available item; TC-Media-06
VA-F-011	The system shall increase volume for a	Functional	High	End User	Volume increases within

Req ID	Requirement	Type	Priority	Source / Stakeholder	Acceptance criteria / Test case
	volume-up command.				maximum; TC-Vol-01
VA-F-012	The system shall decrease volume for a volume-down command.	Functional	High	End User	Volume decreases within minimum; TC-Vol-02
VA-F-013	The system shall mute audio for a mute command.	Functional	Medium	End User	Audio is muted while playback state is retained; TC-Vol-03
VA-F-014	The system shall recognize common equivalent phrases for supported commands.	Functional	High	End User / Developer	Phrase variants map correctly; TC-Intent-02
VA-F-015	The system shall provide feedback when a command is successfully executed.	Functional	Medium	End User	Confirmation is displayed or logged; TC-FB-01
VA-F-016	The system shall provide an error and take no multimedia action for unrecognized or unsupported commands.	Functional	High	End User / Security	No unintended action; TC-Err-01
VA-F-017	The system shall record timestamped command, intent, and execution-result events.	Functional	Medium	Developer / Tester	Required fields logged; raw audio absent; TC-Log-01
VA-F-018	The system shall recover gracefully from speech-recognition or media-player errors where recovery is possible.	Functional	High	Developer / Tester	Application remains usable after injected error; TC-Recovery-01

5. Non-Functional Requirements

Req ID	Requirement	Category	Priority	Acceptance criteria / Measurement
VA-NF-001	Command-to-action response shall be ≤ 3 seconds for at least 90% of recognized commands under normal conditions.	Performance	High	90th percentile ≤ 3 s; TC-Perf-01
VA-NF-002	Intent classification shall achieve at least 90% accuracy for the defined command set in controlled testing.	Accuracy	High	Accuracy $\geq 90\%$; TC-Acc-01
VA-NF-003	The application shall run for 2 continuous hours without an application crash.	Reliability	High	No unplanned crash; TC-Reliability-01
VA-NF-004	Unsupported or low-confidence commands shall not trigger unintended multimedia actions.	Safety / Security	High	100% invalid-command cases produce no unintended action; TC-Sec-03
VA-NF-005	Command feedback shall be understandable and provided within 2 seconds when enabled.	Usability	Medium	$\geq 90\%$ feedback within 2s; TC-UX-01
VA-NF-006	The system shall not retain raw microphone recordings during normal operation.	Privacy / Security	High	No raw audio files after normal run; TC-Privacy-01
VA-NF-007	The voice-control application shall become ready within 30 seconds of application/system startup under the target configuration.	Startup / Performance	Medium	Ready ≤ 30 s in $\geq 90\%$ startup tests; TC-Startup-01

5.1 Security

5.1.1 Security Objectives

SO-01 – Command Integrity: only recognized and supported multimedia intents shall be executed.

SO-02 – Privacy: minimize collection and retention of voice data and do not store raw audio by default.

SO-03 – System Protection: prevent unauthorized modification of command mappings and arbitrary system-command execution.

5.1.2 Security Requirements

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case
VA-SR-001	The system shall execute only intents in the configured multimedia command allow-list.	Security	High	Unsupported intent causes no action; TC-Sec-03
VA-SR-002	The system shall apply a confidence threshold or equivalent validation before executing a voice command.	Security	High	Low-confidence recognition is rejected; TC-Sec-04
VA-SR-003	The system shall not store raw microphone audio unless explicitly enabled for controlled testing.	Privacy	High	No raw audio artifacts in normal run; TC-Privacy-01
VA-SR-004	The system shall prevent recognized text from being executed as arbitrary operating-system commands.	Security	High	Injection-style phrases do not launch OS commands; TC-Sec-05
VA-SR-005	The system shall restrict modification of command mappings and security configuration to authorized administrators.	Security	High	Unauthorized modification is blocked; TC-Sec-06
VA-SR-006	The system shall protect logs	Security	Medium	Permissions/content audit passes; TC-Sec-07

Req ID	Requirement	Type	Priority	Acceptance criteria / Test case
	from unauthorized modification and shall not store passwords, tokens, or raw audio in logs.			

6. Quality Attributes & Acceptance Tests

Exit criteria: all high-priority functional requirements are implemented and verified; no critical NFR/security failures remain; and the RTM shows test coverage.

Voice recognition: TC-Voice-01, TC-Voice-02, TC-Intent-01, TC-Intent-02.

Media: TC-Media-01 through TC-Media-06.

Volume: TC-Vol-01 through TC-Vol-03.

Error/feedback/recovery: TC-FB-01, TC-Err-01, TC-Recovery-01.

Performance/accuracy/reliability/startup: TC-Perf-01, TC-Acc-01, TC-Reliability-01, TC-Startup-01.

Security/privacy: TC-Sec-03 through TC-Sec-07, TC-Privacy-01.

Usability: TC-UX-01.

Acceptance demonstration: the evaluator speaks Play, Pause, Resume, Next, Previous, Volume Up, Volume Down, and Mute commands. The Raspberry Pi shall recognize, validate, and execute them through the media player without unintended actions.

7. UML Use-Case Diagram

The following two use-case views describe the voice assisted multimedia controlling system.

7.1 Use-Case Diagram - Voice Command to Media Control

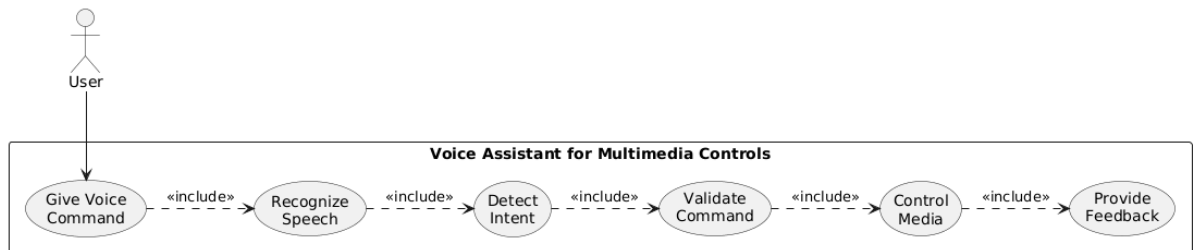


Figure 7.1: Overall, Voice Assistant Use-Case Diagram

7.2 Use-Case Diagram - Multimedia Operations

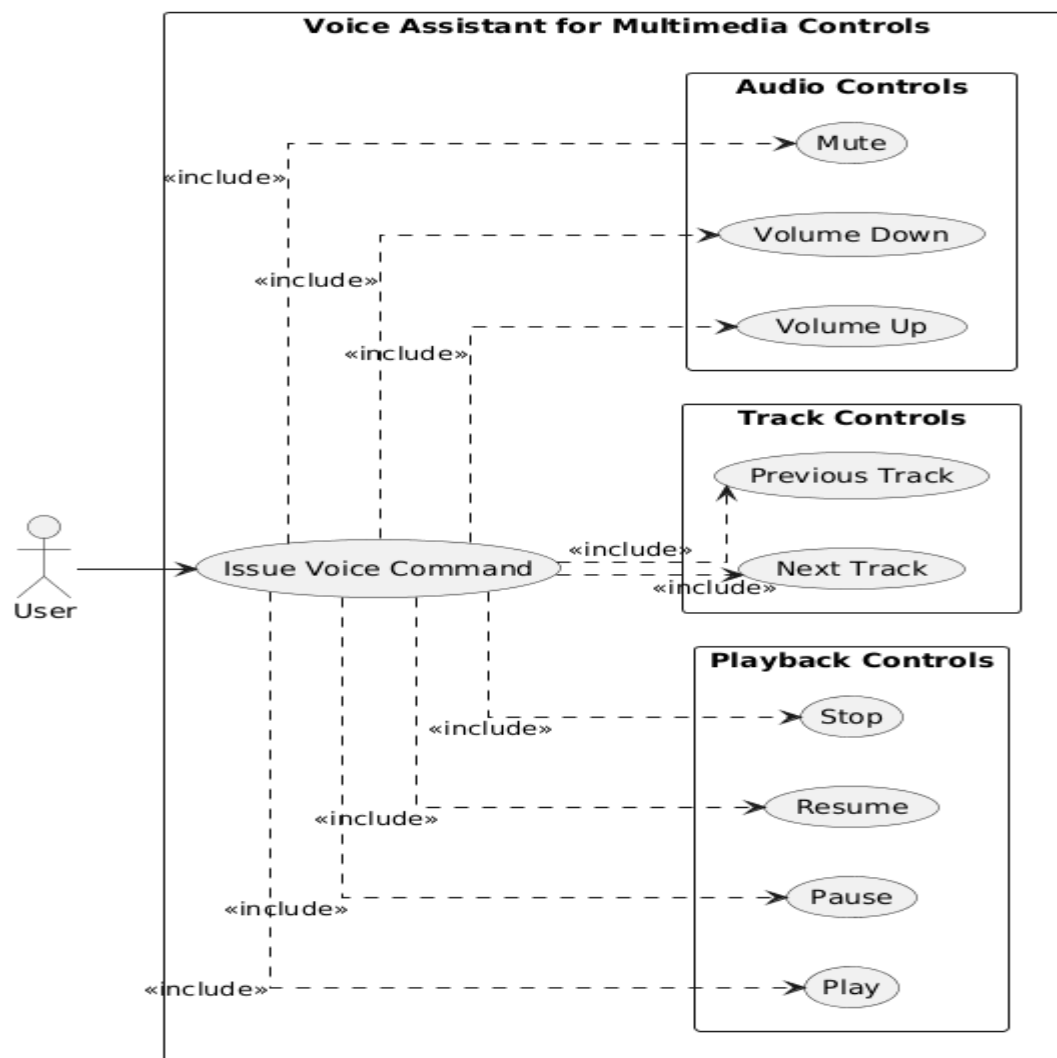


Figure 7.2: Multimedia Control Use-Case Diagram

8. Requirements Traceability Matrix (RTM)

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
V A-F-001	The system shall capture voice input from the USB microphone.	4 / DS-AudioInput	AudioInput	TC-Voice-01	N	Requires USB microphone and audio input configuration.
V A-F-002	The system shall convert captured speech into text.	4 / DS-SpeechRecognition	SpeechRecognition	TC-Voice-02	N	Depends on the selected speech-recognition engine.
V A-F-003	The system shall detect the intent of a recognized multimedia command.	4 / DS-IntentProcessor	IntentProcessor	TC-Intent-01	N	Requires intent-mapping rules for supported commands.
V A-F-004	The system shall validate detected intents against a supported-command allow-list.	4 / DS-CommandValidator	CommandValidator	TC-Sec-03	N	Uses the configured multimedia command allow-list.
V A-F-005	The system shall play media for a play command.	4 / DS-MediaController	MediaController	TC-Media-01	N	Requires VLC/media player integration.
V A-F-006	The system shall pause current media for a pause command.	4 / DS-MediaController	MediaController	TC-Media-02	N	Requires VLC/media player integration.
V A-F-007	The system shall resume paused media for a resume command.	4 / DS-MediaController	MediaController	TC-Media-03	N	Requires VLC/media player integration.
V A-F-008	The system shall stop current media for a stop command.	4 / DS-MediaController	MediaController	TC-Media-04	N	Requires VLC/media player integration.
V A-F-009	The system shall select the next media item for	4 / DS-MediaController	MediaController	TC-Media-05	N	Depends on the active media playlist.

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
	a next command.					
V A-F-010	The system shall select the previous media item for a previous command.	4 / DS-MediaController	MediaController	TC -Media-06	N	Depends on the active media playlist/history.
V A-F-011	The system shall increase volume for a volume-up command.	4 / DS-VolumeController	VolumeController	TC -Vol-01	N	Requires media-player volume control support.
V A-F-012	The system shall decrease volume for a volume-down command.	4 / DS-VolumeController	VolumeController	TC -Vol-02	N	Requires media-player volume control support.
V A-F-013	The system shall mute audio for a mute command.	4 / DS-VolumeController	VolumeController	TC -Vol-03	N	Requires media-player mute support.
V A-F-014	The system shall recognize common equivalent phrases for supported commands.	4 / DS-IntentProcessor	IntentProcessor	TC -Intent-02	N	Requires predefined equivalent-phrase mappings.
V A-F-015	The system shall provide feedback when a command is successfully executed.	4 / DS-FeedbackModule	FeedbackModule	TC -FB-01	N	Feedback may be displayed in terminal/UI and/or logged.
V A-F-016	The system shall provide an error and take no multimedia action for unrecognized or unsupported commands.	4 / DS-ErrorHandler	ErrorHandler	TC -Err-01	N	Must fail safely without changing media state.
V A-F-017	The system shall record timestamped command, intent, and	4 / DS-LoggingModule	LoggingModule	TC -Log-01	N	Logs command, intent, timestamp, and execution

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
	execution-result events.					result; raw audio excluded.
V A-F-018	The system shall recover gracefully from speech-recognition or media-player errors where recovery is possible.	4 / DS-RecoveryManager	RecoveryManager	TC - Recovery-01	N	Requires exception handling and recovery logic.
V A-NF-001	Command-to-action response shall be ≤ 3 seconds for at least 90% of recognized commands under normal conditions.	5 / DS-PerformanceMonitor	PerformanceMonitor	TC -Perf-01	N	Measured from command capture/recognition to media action.
V A-NF-002	Intent classification shall achieve at least 90% accuracy for the defined command set in controlled testing.	5 / DS-AccuracyTest	AccuracyTest	TC -Acc-01	N	Measured using a controlled test set of supported commands.
V A-NF-003	The application shall run for 2 continuous hours without an application crash.	5 / DS-ReliabilityManager	ReliabilityManager	TC - Reliability-01	N	Requires continuous execution/soak testing.
V A-NF-004	Unsupported or low-confidence commands shall not trigger unintended multimedia actions.	5 / DS-CommandValidator	CommandValidator	TC -Sec-03	N	Security and safety validation requirement.
V A-NF-005	Command feedback shall be understandable and provided within 2 seconds when enabled.	5 / DS-FeedbackModule	FeedbackModule	TC -UX-01	N	Depends on the selected feedback mechanism.
V A-NF-006	The system shall not retain raw	5 / DS-PrivacyConfig	PrivacyConfig	TC -	N	Raw audio retention must remain

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
	microphone recordings during normal operation.			Privacy-01		disabled during normal operation.
V A-NF-007	The voice-control application shall become ready within 30 seconds of application/system startup under the target configuration.	5 / DS-StartupManager	StartupManager	TC-Startup-01	N	Measured from application/system startup to ready state.
V A-SR-001	The system shall execute only intents in the configured multimedia command allow-list.	5.1 / DS-CommandValidator	CommandValidator	TC-Sec-03	N	Enforced by command validation/allow-list logic.
V A-SR-002	The system shall apply a confidence threshold or equivalent validation before executing a voice command.	5.1 / DS-ConfidenceValidator	ConfidenceValidator	TC-Sec-04	N	Requires a configured recognition-confidence threshold or equivalent rule.
V A-SR-003	The system shall not store raw microphone audio unless explicitly enabled for controlled testing.	5.1 / DS-PrivacyConfig	PrivacyConfig	TC-Privacy-01	N	Privacy requirement; raw audio disabled by default.
V A-SR-004	The system shall prevent recognized text from being executed as arbitrary operating-system commands.	5.1 / DS-CommandSandbox	CommandSandbox	TC-Sec-05	N	Command input must never be passed directly to the operating system.
V A-SR-005	The system shall restrict modification of	5.1 / DS-ConfigAccessControl	ConfigAccessControl	TC-Sec-06	N	Configuration changes restricted to

Req ID	Requirement short	Section ref / Design Spec	Module	Test case(s)	Status (N/P/A)	Comments
	command mappings and security configuration to authorized administrators.					authorized administrators.
V A-SR-006	The system shall protect logs from unauthorized modification and shall not store passwords, tokens, or raw audio in logs.	5.1 / DS-SecureLogging	SecureLogging	TC-Sec-07	N	Logs must contain no passwords, tokens, or raw audio and should be protected from unauthorized modification.

Division of Work

YASH RAGHAVENDRA A: Prepared the system requirements, overall system design, architecture, UML use-case modelling, and integrated the final SRS documentation.

KUSHI NIRANJANA: Worked on speech recognition, speech-to-text processing, intent detection, voice-command mapping, and command validation.

HEMANT RAM NELAPATI: Handled Raspberry Pi setup, USB microphone and audio configuration, VLC/media-player integration, and hardware-software integration.

KEERTHI B PATIL: Prepared test cases, security requirements, non-functional requirement validation, error handling tests, and maintained the Requirements Traceability Matrix.